Techno-Economic and Environmental Pre-Assessment of Large-Scale Heat Pumps Using the HeatSHIFT Webtool

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Abstract:
Heat pumps are expected to play a central role in the decarbonization of the heating sector. Despite their technical potential to cover Germany's entire heat demand up to 200 °C, the market uptake – particularly for large-scale applications – remains slow. Complex planning and approval procedures often hinder early project development. To address this challenge, the *HeatSHIFT* project has developed a web-based preliminary design tool that enables the systematic assessment and comparison of large-scale heat pump systems. The tool incorporates technical, economic, and environmental evaluations and requires minimal user input by providing representative default values and simplified equations. It includes refrigerant selection, process simulation, cost estimation, subsidy modeling, and scenario as well as sensitivity analyses.
Complementing this, an interactive, GIS-based platform for waste heat mapping and analysis has been developed to identify, visualize, and evaluate potential industrial waste heat sources across Germany. The platform builds upon data provided by the German Federal Office for Energy Efficiency's Waste Heat Platform (BAFA/BfEE Plattform für Abwärme - PfA) and allows users to apply spatial and technical filters, assess source temperature levels and availability, and examine proximity to existing or planned district heating networks.
Together, these tools support planners, utilities, and municipal stakeholders in making data-driven decisions and accelerating the deployment of large-scale heat pumps in district heating systems.

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Contents

1 Introduction **4**

2 *HeatSHIFT* research project **4**

3 *HeatSHIFT Webtool* **5**
3.1 Disclaimer 5
3.2 Technological evaluation 5
3.2.1 Refrigerant selection 5
3.2.2 Cycle simulation 7
3.3 Economic evaluation 7
3.3.1 Scenario analysis 8
3.3.2 Sensitivity analysis 8
3.3.3 Estimation of potential subsidies 9
3.4 Ecological evaluation 10

4 *HeatSHIFT Interactive Platform for Waste Heat Mapping and Analysis* **11**
4.1 Conceptual Framework 11
4.2 Data Sources and Structure 11
4.2.1 Waste Heat Data (BAFA Platform) 11
4.2.2 Heat Demand Areas 12
4.2.3 Categorisation of Waste Heat Sources 12
4.3 Spatial and Functional Analysis 13
4.3.1 Spatial Proximity and Reachability Analysis 14
4.3.2 Mapping and User Interaction 15

5 Conclusions **19**

6 Acknowledgements **19**

Abbreviations

BAFA	Federal Office for Economic Affairs and Export Control	BEW	Federal Funding for Efficient Heating Networks
BfEE	Federal Office for Energy Efficiency	CHP	Combined heat and power
COP	Coefficient of performance	DH	District heating
GWP	Global Warming Potential	KWKG	Combined Heat and Power Act
NPV	Net present value	ODP	Ozone Depletion Potential
PfA	Waste Heat Platform	SCOP	Seasonal coefficient of performance

Nomenclature

$\dot{Q}$	Thermal load (kW)	P	Power (kW)
T	Temperature (K)	η_g	Second law efficiency

1. Introduction

Heat pumps are considered a key technology for the energy transition. According to [1], heat pumps are technically capable of meeting the entire heat demand in Germany up to temperatures of 200 °C. This demand currently accounts for over 75% of the national natural gas consumption. As such, heat pumps represent a central pillar in the decarbonization of the heating sector. However, [1] also highlights that the market ramp-up – especially for large-scale heat pumps – remains slow. This is primarily due to complex planning and permitting procedures.

As part of the research project *HeatSHIFT*[1,2], a preliminary design tool has been developed to support the early-stage assessment and comparison of large-scale heat pump systems. In the following sections, the *HeatSHIFT* project is briefly introduced, and the structure, functionality, and application of the tool are presented in detail.

In addition to the development of the large-scale heat pump pre-assessment tool, a web-based GIS environment was implemented to systematically explore and evaluate industrial waste heat potentials across Germany. The platform draws upon data from the German Federal Office for Energy Efficiency's Waste Heat Platform [14] and provides advanced mapping and filtering functionalities to identify promising waste heat sources and assess their suitability for district heating integration. By linking geospatial information with technical and operational parameters such as temperature level, availability, and proximity to potential heat consumers, the platform enables a more comprehensive understanding of regional waste heat potentials.

2. *HeatSHIFT* research project

In order to achieve the German governments climate targets for 2045 and the associated extensive decarbonization of heat supply, district heating (DH) plays a key role. Against the backdrop of emission reduction and the simultaneous enhancement of supply security, the research project *HeatSHIFT* aims to investigate efficient integration options for high-temperature heat pumps in DH networks. The analyses focus on existing DH systems with high supply temperatures of 120 °C and above.

Figure 1 provides an overview of the scope and partners of the *HeatSHIFT* research project.

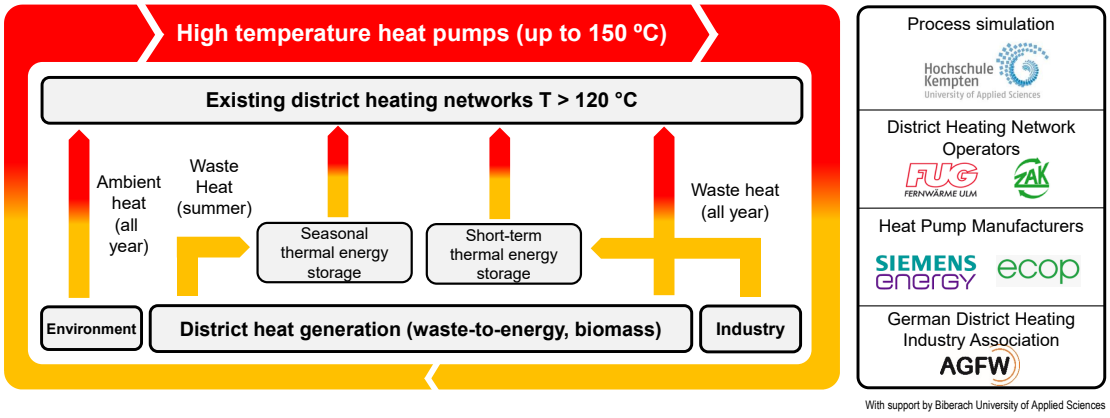


Fig. 1. *HeatSHIFT* research project scope and partners.

The core objective of the project is the systematic investigation and both technical and economic evaluation of the use of high-temperature heat pumps through process simulation, taking into account various heat sources and high-temperature heat pump technologies. To this end, realistic process simulation models of high-temperature heat pumps and combined heat and power (CHP) plants – particularly biomass and waste-to-energy plants – are developed and combined based on data provided by the project

[1]Project Homepage: `https://bayfis.fza.hs-kempten.de/de/forschungsprojekt/482-heatshift`
[2]Funding information: `http://www.enargus.de/detail/?id=23450180`

partners. On the basis of validated process simulation models, the integration of high-temperature heat pumps is optimized, and an economic evaluation is performed.

Furthermore, a simplified tool for the preliminary design of heat pump applications is developed for heat suppliers. This tool is introduced and described in more detail in the following section.

3. *HeatSHIFT Webtool*

The *HeatSHIFT Webtool* aims to facilitate the initial steps in the planning process of large-scale heat pump systems by enabling a systematic comparison of available options and supporting data-driven decision-making. It is intended for planners, utility companies, research institutions, and municipal decision-makers seeking to actively contribute to the transition toward a climate-neutral heat supply.

The *HeatSHIFT Webtool* is available online at: `https://heat-shift.de/`.

It is designed to operate with minimal prior information. To enable its application even in the absence of detailed input data, representative default values and approximate equations have been predefined for all key parameters. These values are derived from current literature, standard assumptions, and official sources. While they provide a useful basis for initial orientation, they should be adjusted accordingly whenever more accurate information becomes available.

The webtool is structured into three main components: technical, economic, and environmental assessment. These components, along with a preceding disclaimer, are presented in the following subsections.

3.1. Disclaimer

The authors explicitly disclaim any liability for the accuracy, completeness, or applicability of the results generated by this tool. The methodological framework and default values are intended to support preliminary assessments of the technical, economic and environmental performance of large-scale heat pump systems. The results are not designed to replace professional engineering judgment or detailed project-specific planning. Users are therefore advised to validate the results with site-specific data and consult relevant experts before drawing any conclusions or making investment decisions.

3.2. Technological evaluation

To assess the technical feasibility of integrating a heat pump into a heating system, the tool first determines suitable refrigerants based on the specified temperature levels. Subsequently, a heat pump is simulated using the refrigerant selected by the user. Key performance indicators and process variables are provided as outputs. The only required inputs are the inlet and outlet temperatures of the heat source and heat sink, as well as either the required useful heat output or the available thermal power from the heat source. The user may choose which of the two thermal power parameters to specify.

3.2.1. Refrigerant selection

The tool considers the following refrigerants: R32, R1270, R290, R717, R1234yf, R134a, R227ea, R600a, R236fa, R600, R718, R245fa, R1233zd(E), R245ca, and R601. To define the applicable temperature range for each refrigerant, the lower temperature boundary is set to 5 K above the boiling temperature at a pressure of 1 bar. For R718, an exception is made due to its thermophysical properties, and a reference pressure of 611.655 Pa is applied. The upper temperature boundary is defined as 15 K below the respective critical temperature [2]. Thermophysical property data are retrieved from the *CoolProp* database [3]. The resulting temperature ranges are summarized in Table 1.

Table 1. Applicable temperature ranges for selected refrigerants.

Refrigerant	**Lower temperature limit [°C]**	**Upper temperature limit [°C]**
R32	−46,91	66,11
R1270	−42,91	76,06
R290	−37,41	81,74
R717	−28,58	117,41
R1234yf	−24,78	79,70
R134a	−21,36	86,06
R227ea	−11,65	86,75
R600a	−7,08	119,67
R236fa	3,20	109,92
R600	4,16	136,98
R718	5,01	358,95
R245fa	19,72	138,86
R1233zd(E)	22,91	151,45
R245ca	29,92	159,42
R601	40,67	181,55

From the set of refrigerants suitable for the specified operating temperature range, the user selects one for subsequent process simulation, as illustrated in Figure 2.

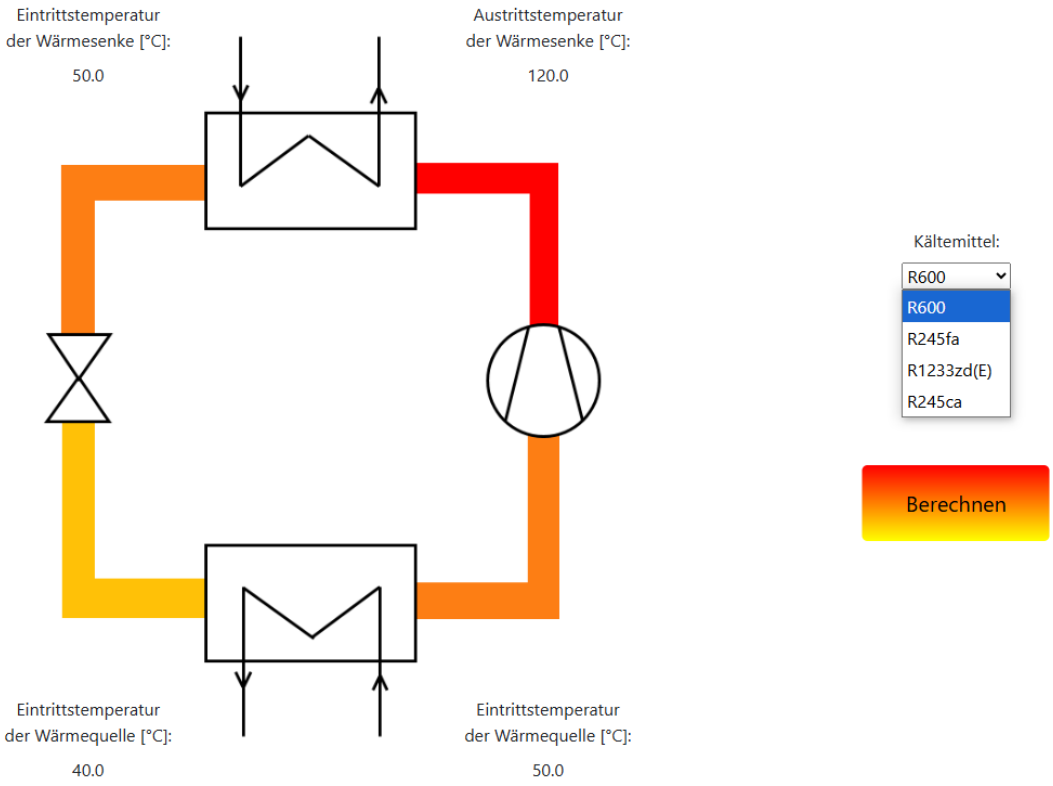


Fig. 2. Example selection of refrigerants matching the operating temperature range.

3.2.2. Cycle simulation

Based on the selected refrigerant, the thermodynamic cycle of a single-stage vapor-compression heat pump is subsequently simulated. A terminal temperature difference of 5 K is assumed for both heat exchangers, and a superheat of 5 K is specified at the compressor outlet. The isentropic compressor efficiency is assumed to be 85 %, while the mechanical efficiency is set to 95 %.

Depending on the user input – either the useful heat output $\dot{Q}_{\text{sink}}$ or the heat input from the source $\dot{Q}_{\text{source}}$ – the respective counterpart, as well as the compressor's electrical power consumption P_{el}, is determined using thermophysical property data from *CoolProp*. The coefficient of performance (COP) of the heat pump is subsequently calculated according to Equation 1.

$$\text{COP} = \frac{\dot{Q}_{sink}}{P_{el}} \tag{1}$$

Furthermore, the theoretical Carnot COP and the second law efficiency η_g of the heat pump are computed using Equations 2 and 3, respectively. These metrics serve to evaluate the thermodynamic quality of the process. For the calculation of the Carnot COP, the boiling temperatures $T_{boil,up}$ and $T_{boil,low}$ corresponding to the respective pressure levels are applied.

$$\text{COP}_{\text{Carnot}} = \frac{T_{boil,up}}{T_{boil,up} - T_{boil,low}} \tag{2}$$

$$\eta_g = \frac{\text{COP}}{\text{COP}_{\text{Carnot}}} \tag{3}$$

In addition, the mass flow rates of water on both the source and sink sides are determined to quantify the thermal exchange at the respective temperature levels. Finally, the process is visualized in a *p-h* diagram, as shown in Figure 3.

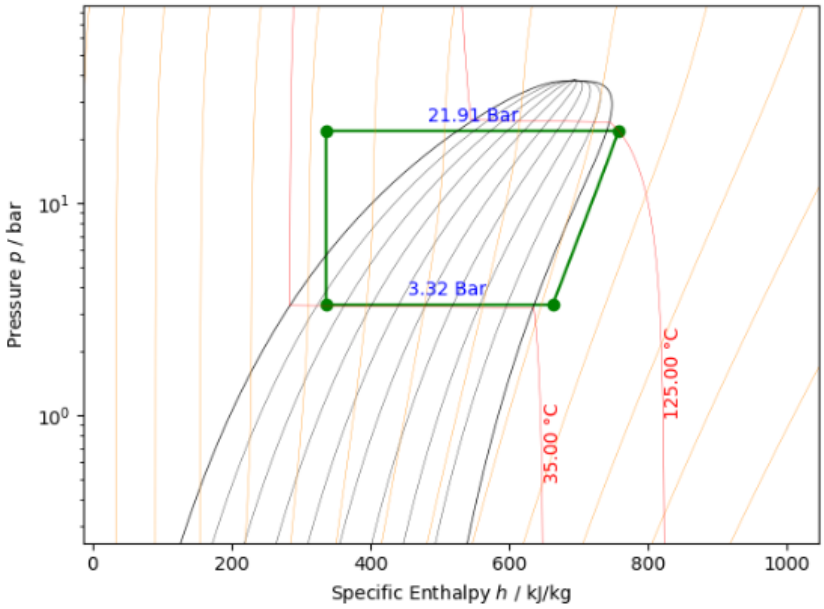


Fig. 3. Visualization of a representative process in a *p-h* diagram.

3.3. Economic evaluation

The economic evaluation requires a range of input parameters. In addition to the investment and operating costs of the heat pump system, system- and operation-specific parameters must be provided, such as the nominal capacity, COP, annual full-load hours, type of heat source, and expected service life. If investment and operating costs are not available, they are estimated based on [4] and [5].

Furthermore, economic indicators such as the interest rate, inflation rate, and energy price fluctuations are also of relevance. The electricity price for the purchased electricity must be specified. It can also be indicated whether industrial electricity tariffs apply. Default values are based on [6]. Any share of electricity from self-generation is not included in the electricity cost calculation.

The economic benefit is defined as the avoided cost of natural gas. Therefore, the gas price must also be provided. The default value is likewise taken from [6].

Finally, it can be specified whether the heat pump is to be integrated into a CHP system. If so, financial support under the Combined Heat and Power Act (KWKG) may be available. In this case, the electrical and thermal output of the CHP system must also be specified.

The economic evaluation comprises not only the calculation of levelized cost of heat but also includes a scenario and sensitivity analysis as well as an option to estimate potential subsidies.

3.3.1. Scenario analysis

The scenario analysis is conducted in accordance with DIN 17463. The standard recommends a parameter variation of 50%, unless other values are known [7]. The parameters subjected to variation include the annual useful energy, the investment lifetime, investment costs, discount rate, operating costs excluding energy costs, electricity costs, avoided gas costs, share of self-generated electricity, inflation, and energy price volatility.

The resulting key figures are the net present value (NPV) and the payback period. The development of the NPV is also illustrated graphically as examplarily shown in Figure 4 as well as in tabular form.

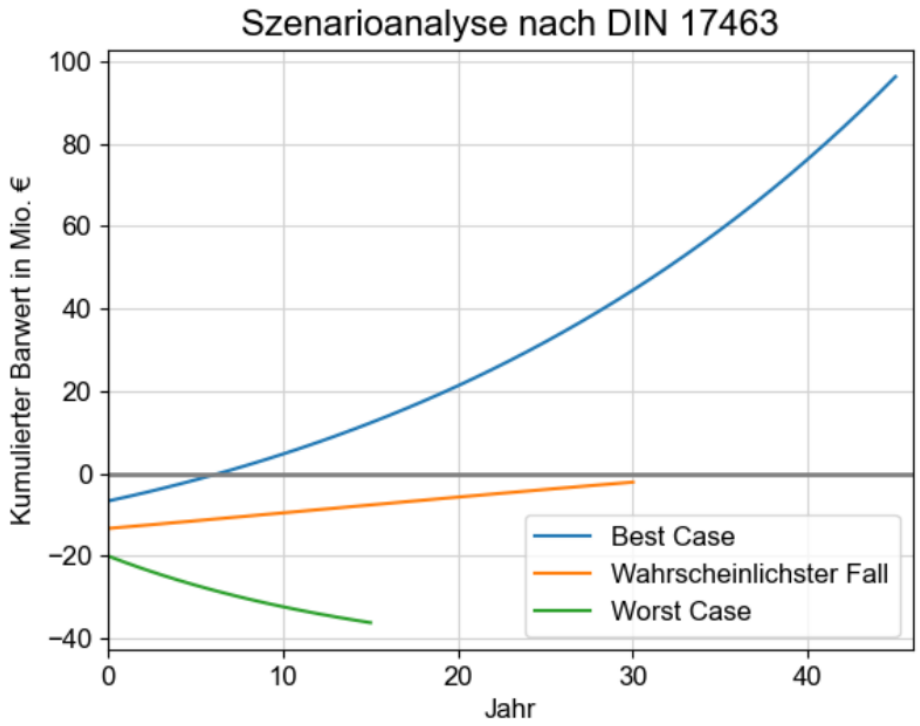


Fig. 4. Scenario analysis according to DIN 17463.

3.3.2. Sensitivity analysis

The sensitivity analysis is likewise conducted in accordance with DIN 17463. For this purpose, the parameters annual useful energy, investment lifetime, investment costs, discount rate, operating costs excluding energy costs, electricity costs, gas costs, inflation, and energy price volatility are each varied by 50%. The impact on the NPV is then evaluated. The results are presented both graphically as exemplarily shown in Figure 5 and in tabular form.

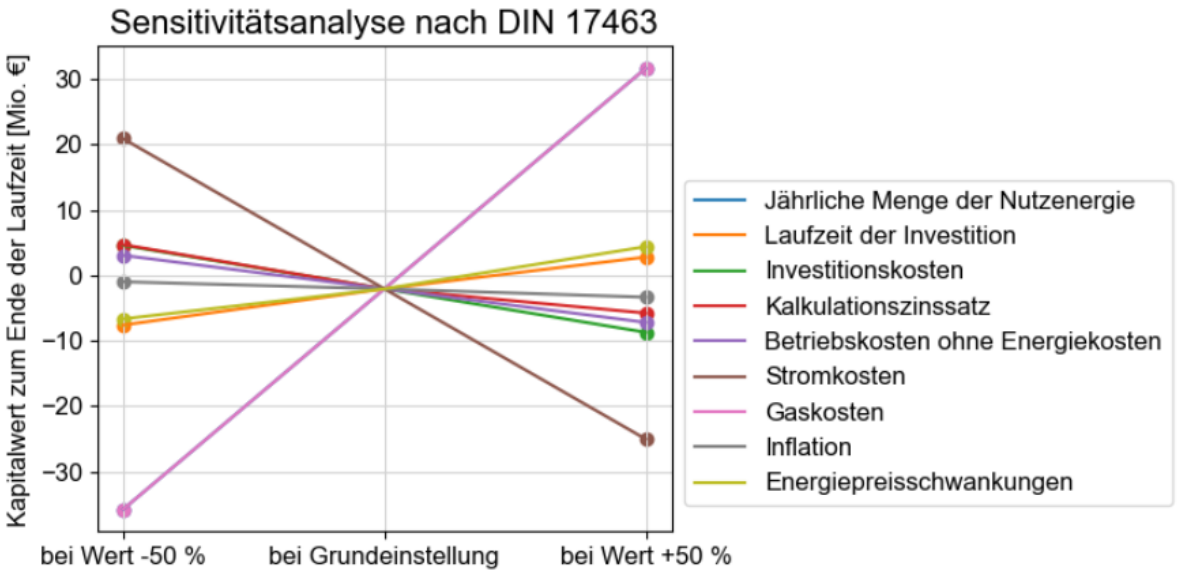


Fig. 5. Sensitivity analysis according to DIN 17463.

3.3.3. Estimation of potential subsidies

The economic viability of plant operation can be significantly improved through the use of available subsidies. Two potential funding mechanisms are considered in this context. Firstly, a subsidy from the Federal Office for Economic Affairs and Export Control (BAFA) under the Federal Funding for Efficient Heating Networks (BEW funding) may be applicable [8]. Additionally, if the heat pump is part of an innovative CHP system, surcharge payments for the generated CHP electricity can be applied for under the KWKG [9].

The BEW funding options comprise four modules. For heat pump funding, Modules 2 or 3 and Module 4 are relevant. Modules 2 or 3 cover the heat pump as an individual measure and support associated investment costs, while Module 4 serves to subsidize operating costs. Funding under Modules 2 or 3 covers 40% of the expenses that can be clearly attributed to the individual measure and is capped at the economic viability gap or €100,000,000 [8]. The economic viability gap is defined as the NPV over 30 years of operation, assuming an inflation rate of 2% and a discount rate of 6% [5]. The funding level under Module 4 depends on the type of electricity generation. Maximum subsidy rates vary depending on whether the electricity is drawn from the grid or generated from a renewable energy source. In the case of mixed electricity sourcing, the subsidy amounts are calculated proportionally. The operating cost subsidy is limited to 90% of electricity costs. An application may be submitted for a maximum of €100,000,000 or the economic viability gap. A minimum efficiency of the system must be demonstrated through a seasonal coefficient of performance (SCOP) of at least 2.5. The funding is granted for a period of 10 years and is disbursed annually [8].

To obtain funding for the use of a heat pump within the framework of an innovative CHP system under the KWKG, several requirements must be fulfilled. The entire system must consist of a CHP plant with an electrical output of more than 10 MW, combined with a heat pump and an additional electric heat generator operated under a joint control system. These heat generators must feed into a common heat network. Furthermore, the heat pump must be powered by electricity or a gaseous medium and must achieve an annual performance factor of at least 1.25 [9]. The specific requirements are highly complex and must be examined in detail on a case-by-case basis, including the level of surcharge payments. The surcharge is granted as an additional funding component for the generated CHP electricity. The eligible subsidy amount is determined based on the share of innovative heat relative to the reference heat. The reference heat is defined as the useful heat of the CHP system at 3,000 full load hours per year combined with the useful heat of the heat pump [9]. Table 2 shows the surcharge in accordance with KWKG § 7a.

Table 2. Surcharges pursuant to KWKG § 7a for innovative renewable heat.

Share of innovative renewable heat [%]	**Surcharge [ct/kWh CHP electricity]**
≥ 5 %	0.4
≥ 10 %	0.8
≥ 15 %	1.2
≥ 20 %	1.8
≥ 25 %	2.3
≥ 30 %	3.0
≥ 35 %	3.8
≥ 40 %	4.7
≥ 45 %	5.7
≥ 50 %	7.0

3.4. Ecological evaluation

To provide an initial estimate of the environmental impact of the potential heat pump system, various indicators are determined under the section for ecological assessment. In addition to the nominal capacity and COP of the heat pump, the user must specify the annual full-load hours, service life, and selected refrigerant. Moreover, the electricity generation source used to operate the heat pump must also be selected.

The calculation determines the amount of CO_2 emissions that can be avoided annually and over the entire lifetime of the system through the deployment of a large-scale heat pump. For comparison, a reference scenario is considered in which the same amount of useful heat is provided by a natural gas-fired boiler with an assumed boiler efficiency of 95 %. The emission factor and calorific value of natural gas are taken from [10], while the emission factors for various electricity generation technologies are based on [11] and [12].

In addition, information on the selected refrigerant – such as its safety classification, atmospheric lifetime, Global Warming Potential (GWP), and Ozone Depletion Potential (ODP) [2] – is provided to facilitate a more comprehensive understanding of the refrigerant-related environmental impact.

4. *HeatSHIFT Interactive Platform for Waste Heat Mapping and Analysis*

This additional tool enables users to identify, visualize, and analyze industrial waste heat sources and their proximity to high heat demand density areas in Germany. By integrating publicly available data from the German Federal Office for Energy Efficiency's Waste Heat Platform (BAFA PfA) and combining it with additional geospatial datasets, the tool supports stakeholders in recognizing technically and economically feasible recovery options.

The following chapters present the conceptual framework, data foundations, software architecture, and functional capabilities of the platform.

The *HeatSHIFT Interactive Platform for Waste Heat Mapping and Analysis* is not integrated into the website `https://heat-shift.de/`, but is provided as Python code via a GitHub repository available at `https://github.com/HS-Kempten/PfA_Map`.

4.1. Conceptual Framework

This platform is designed as a decision-support and exploration tool that bridges the gap between raw data and practical planning processes. All displayed information is based on publicly accessible or open data, ensuring reproducibility and accountability. A web-based, map-centered interface enables users to intuitively explore complex datasets without requiring GIS or data analytics expertise. In essence, the platform aims to help users identify which waste heat sources are particularly suitable for utilization.

Rather than performing detailed techno-economic calculations, the platform provides a screening-level assessment that helps users identify and prioritize potential projects for further study. Determining optimal options for excess heat utilization is naturally complex, as it involves balancing technical feasibility, economic viability, environmental impacts, regulatory constraints, and the specific objectives of each organization [22]. Given this complexity, the platform deliberately focuses on providing an initial, transparent screening rather than a full decision-support model. While a formal evaluation matrix or scoring methodology is not yet implemented, its development is intended as part of future research.

The following parameters represent the key technical and spatial criteria for user-driven assessments of waste heat utilization potential, covered by the presented platform.

1. **Temperature level** – determines whether the waste heat can be directly fed into a network or requires upgrading via a heat pump.
2. **Available thermal power (heat quantity)** – indicates the potential contribution to local heat demand.
3. **Temporal availability** – shows how well the source aligns with seasonal heat demand.
4. **Type of waste heat (medium)** – such as cooling water, exhaust air, or process water. This information is derived through additional classification steps unique to the developed tool.
5. **Spatial proximity to potential heat consumers** – evaluated through geospatial analysis of distances between waste heat sources (WHS) and district heating areas (DHA).

The first three parameters originate from the BAFA dataset, whereas the last two - waste heat type and spatial proximity - represent distinct features of this platform and significantly enhance its analytical value.

4.2. Data Sources and Structure

4.2.1. Waste Heat Data (BAFA Platform)

The *Waste Heat Platform* (Plattform für Abwärme, PfA) of the German Federal Office for Energy Efficiency (BAFA) is the primary data foundation of the webtool and constitutes one of the most comprehensive national datasets on industrial waste heat potentials [14]. As of the most recent update, it

comprises information from approximately 3,000 companies and about 26,000 individual waste heat sources, corresponding to a total reported annual waste heat potential of around 244 TWh.

This national database contains information reported by industrial and commercial facilities across Germany. According to the underlying legal framework, only installations exceeding specific annual final energy consumption thresholds are required to report, while smaller facilities are exempt. Furthermore, data publication may be restricted upon request if confidentiality concerns exist. The dataset includes company and site information such as street addresses, temperature ranges, available thermal power, control options and indicative operational periods. Exact geographic coordinates are not provided.

Overall, the BAFA data offer a valuable national overview of industrial waste heat potentials but remain heterogeneous in quality and completeness. In particular, attributes such as the type of heat carrier medium or detailed temporal availability profiles are not included, which limits their direct applicability for project-level analyses.

4.2.2. Heat Demand Areas

To contextualize these sources within their regional energy infrastructure, this webtool integrates additional spatial data representing areas with existing or expected district heating demand. Ideally, such information would be based on geospatial datasets of existing district heating networks, as provided by several regional mapping initiatives, e.g. the Energieatlas NRW) [16]. However, comprehensive nationwide coverage of district heating network geodata is currently not available.

As an alternative, the *Pan-European Thermal Atlas (PETA 5.2)* developed within the *Heat Roadmap Europe* and *sEEnergies* projects is used [17, 21, 25]. PETA5.2 provides a harmonized European dataset of heat demand densities, categorizing areas into five demand classes in gigajoules per hectare (GJ/ha). For the present analysis, density classes 3 to 5 (ranging from approximately 500 to 3,000 GJ/ha) are selected, representing typical urban and suburban zones where district heating systems are either already present or technically and economically viable. The data is provided through the *Halmstad University District Heating and Cooling Database* and serves as an indicator for both existing and anticipated district heating areas.

By combining industrial waste heat data from the BAFA platform with regional heat demand density information from PETA, the webtool enables users to identify spatial correlations between potential heat sources and likely consumer areas. This coupling forms the foundation for subsequent proximity analyses and qualitative suitability assessments.

4.2.3. Categorisation of Waste Heat Sources

Since the BAFA dataset does not explicitly specify the physical heat carrier or the underlying process type of each waste heat source, an additional categorisation step was implemented to infer this information from the available textual fields. Several classification approaches were evaluated, including a keyword-based rule model, large language models (LLMs), and conventional machine learning algorithms.

All methods were trained and validated on a manually labelled subset of roughly 1,000 entries. Among them, the keyword-based approach delivered the most robust and interpretable performance while ensuring full transparency for users. It was therefore selected for the final workflow.

To enable a more detailed technical interpretation of the reported waste heat potentials, the classified sources were grouped into eight categories. These categories represent typical physical media or process types from which waste heat arises and therefore support the assessment of possible recovery technologies. The keyword-based classification was performed using German-language input terms, as these reflect the original entries in the BAFA database. Table 3 provides an overview of the applied classification scheme, including concise descriptions and representative keywords used for assignment.

Table 3. Categorisation of waste heat sources based on keyword analysis.

Category	Description	Typical Keywords
Cooling Water	Waste heat discharged through cooling circuits, condensers, or cooling towers. Common in power generation, metal processing, and chemical industries.	cooling water, condenser, cooling tower, heat exchanger
Wastewater	Thermal energy contained in process effluents or sewage flows. Relevant for food industry, paper production, or wastewater treatment plants.	wastewater, process water, sewage, treatment plant
Exhaust Air / Flue Gas	Hot exhaust gases or air from combustion, drying, or ventilation processes. Frequently found in ovens, kilns, and thermal dryers.	exhaust air, flue gas, chimney, furnace, combustion
Cooling / Refrigeration System	Heat rejected by refrigeration or air-conditioning systems, typically at low to medium temperature levels. Found in food processing, cold storage, and HVAC installations.	refrigeration, chiller, cooling unit, ice water, air conditioning
CHP / Power Plant / Turbine	Waste heat from combined heat and power units, turbines, or engines, partly already recovered in existing systems.	CHP, turbine, engine, generator, power plant
Compressed Air / Compressor / Pump	Heat generated by mechanical compression processes, such as compressors, pumps, or air systems. Often continuous and available at moderate temperatures.	compressor, pump, compressed air, blower, pressurised air
Electronics / Transformer	Low-grade waste heat emitted by electrical equipment such as server farms, transformers, converters, or power electronics.	transformer, converter, rectifier, inverter
Other / Undefined	Entries with insufficient information or mixed characteristics that cannot be assigned clearly.	—

Classifying the waste heat sources provides several important benefits for both data interpretation and practical decision-making. It transforms the heterogeneous and often unstructured textual information in the BAFA dataset into a consistent, analysable format. This enables a more systematic comparison of sources and supports meaningful aggregations across different sectors and technologies. Identifying the underlying physical medium or process type is essential for assessing which recovery technologies may be technically feasible, since suitable heat exchangers, hydraulic concepts, or heat pump configurations strongly depend on the nature of the waste heat stream. Additionally, the categorization enhances the usability of the platform by allowing users to filter, interpret, and prioritize waste heat potentials according to their specific planning objectives.

4.3. Spatial and Functional Analysis

Beyond the integration and classification of waste heat data, the webtool provides functionalities that enable spatial interpretation and visual interaction with the processed information. These features are designed to help users explore the spatial relationships between industrial waste heat sources and potential heat consumers and to assess the local feasibility of heat recovery and integration into district heating systems. The analysis focuses on two main components: the determination of spatial proximity between waste heat sources and heat demand areas, and the map-based user interface that visualizes these relationships through an intuitive and interactive representation.

To operationalize the conceptual approach described above, the platform integrates several geospatial

and analytical components. Road network data were retrieved via the OSMnx Python library, which accesses OpenStreetMap data [15, 23]. Network analysis was performed using NetworkX [19]. Because the BAFA PfA does not provide exact geographic coordinates, these were initially derived through automated geocoding using the OSM Nominatim service.

4.3.1. Spatial Proximity and Reachability Analysis

To assess the spatial relationship between industrial waste heat sources (WHS) and potential consumer areas, a polygon-based proximity and reachability analysis was implemented. This approach allows users to identify which heat sources are located within a technically or economically reasonable distance from existing or expected district heating regions.

The analysis combines two complementary spatial concepts:

1. **Linear proximity:** The straight-line distance between each waste heat source and the nearest district heating area (DHA) border is computed to provide a general indication of spatial closeness. This metric is computationally efficient and serves as an initial screening parameter.

2. **Network-based reachability:** In order to represent realistic connection paths, a second analysis step determines the reachable area along existing transport and infrastructure networks. The calculation is based on road and pathway data from OSM, which are converted into a network graph representation. For each waste heat source, the algorithm identifies the nearest accessible node within the OSM network. Since many industrial facilities are located on large properties with internal roads or courtyards not represented in OSM, the search radius around the source coordinate is dynamically increased until a valid connection to the public network is established.

   From this starting node, all possible paths along the network are explored using a Dijkstra shortest-path algorithm, which computes the cumulative travel distance along connected edges, as provided by the NetworkX implementation [19]. All end-nodes reachable within the defined maximum path length of 500 m are stored, collectively defining the reachability area of the waste heat source. The set of reachable end points is then converted into a polygon using convex hull and smoothing operations to delineate the reachability area around the source. This resulting polygon visually represents the potential spatial extent within which a technical heat connection could be established (see Fig. 6).

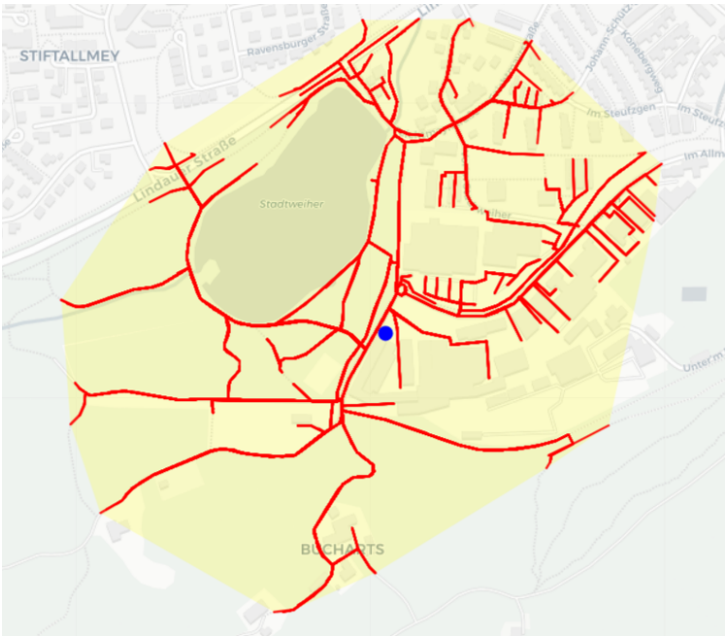


Fig. 6. Visualization of a representative WHS reachability area.

The threshold value (500 m) was chosen as a heuristic parameter to generate representative connection ranges while maintaining computational efficiency. It does not imply an economic optimum but serves as an illustrative reference distance for the visualization of technically accessible areas. In practice, however, the feasible connection distance of a district heating pipeline is influenced by several factors. These include the specific investment cost per meter of pipeline construction (covering materials, excavation, and surface reinstatement), the heat demand density along the potential route and the resulting annual heat utilization, as well as the expected thermal losses and their effect on overall system efficiency. Moreover, the available connection power and the temperature level of the waste heat source play an important role. Finally, topographical and infrastructural constraints, such as terrain characteristics, necessary crossings, or the availability of suitable road corridors, can significantly affect the practicality and cost of a pipeline connection [18].

The resulting polygons are intersected with the heat demand layer (from PETA) to reveal spatial overlaps between potential sources and consumer areas, as illustrated in Fig. 7. This intersection analysis provides a first-order indicator of technical connection opportunities. It allows users to visually explore how many and which sources are located within the feasible connection range of heat demand clusters.

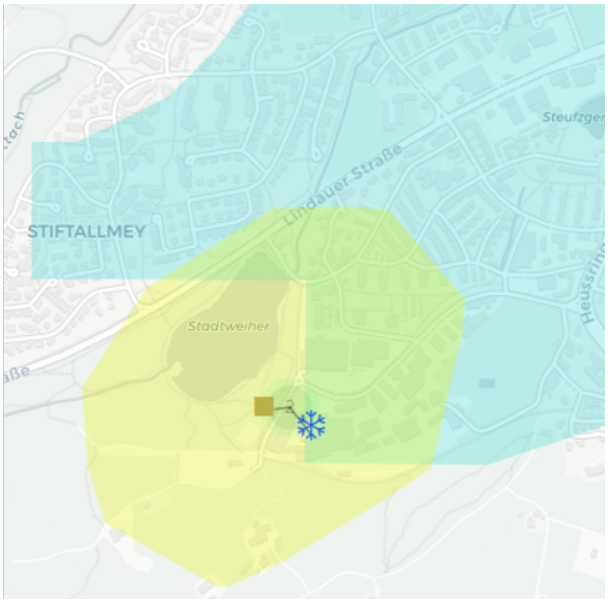


Fig. 7. Intersection of WHS reachability area and DHA.

4.3.2. Mapping and User Interaction

The visual interface of the webtool is designed to provide an intuitive and transparent representation of industrial waste heat potentials and their spatial context. It enables users to interactively explore, filter, and interpret geospatial data through a web-based map environment built upon the *Folium* and *Leaflet* libraries [20]. All map elements are generated from the processed geospatial dataset.

Filtering Functionality. The webtool includes an integrated filtering interface (see Fig. 8) that allows users to tailor the displayed dataset according to technical and spatial criteria.

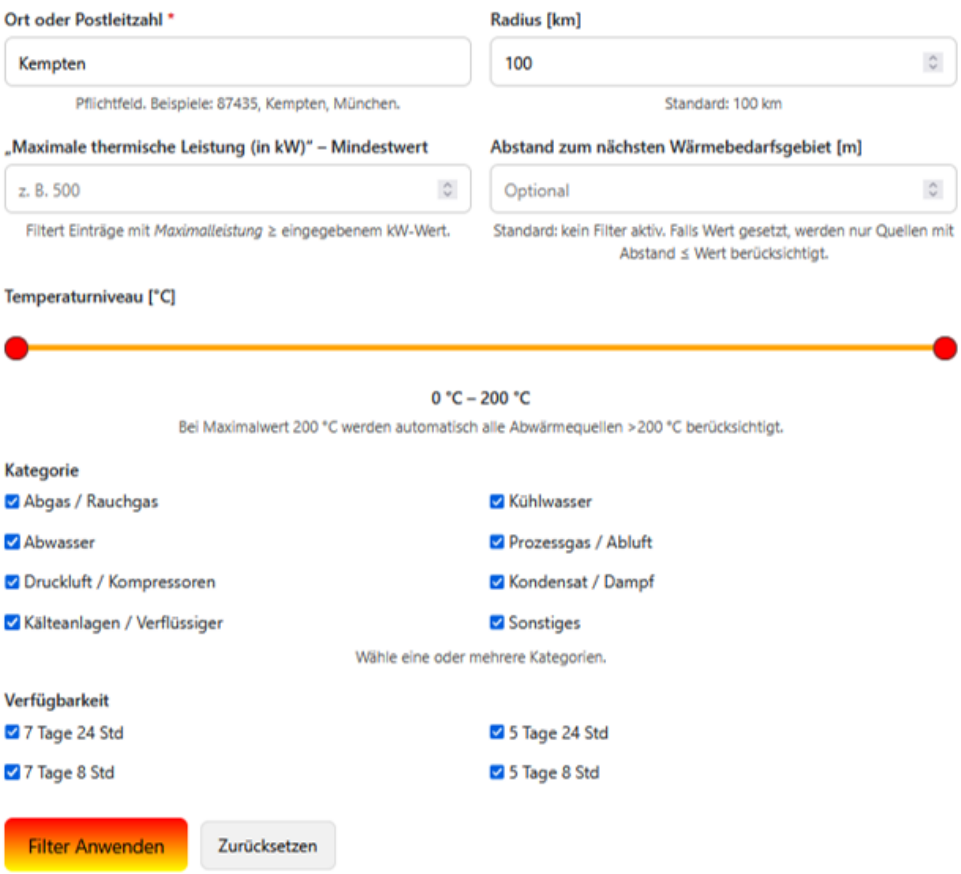


Fig. 8. Filtering interface.

Filters can be applied to specific regions, performance ranges, temperature levels, and waste heat categories. Users may enter a city name or postal code to define a search centre and specify a search radius in kilometres. Additional parameters include a minimum available thermal power and optionally a maximum distance to the nearest district heating area, enabling focused analysis of spatially relevant sources.

A temperature slider allows dynamic restriction of the displayed sources to selected temperature intervals between 0 °C and 200 °C. Further checkboxes enable filtering by waste heat category as described in Table 3 and by indicative availability periods (e.g., continuous operation or weekday schedules). Once applied, the filters immediately update the map view, allowing users to refine their search and focus on technically and spatially feasible heat sources within the defined area of interest.

Map Layout and Layer Structure. The central element of the webtool is an interactive map with adjustable zoom levels. At the national scale, users obtain an overview of the spatial distribution of waste heat sources across Germany, while regional zoom levels allow for detailed inspection of individual sites and their surroundings. Separate layers are provided for:

- Industrial waste heat sources (WHS),
- District heating or high heat demand areas (DHA),
- Network-based reachability areas of individual sources,
- Spatial intersections between WHS and DHA polygons.

Visual Encoding and Symbol Design. Each waste heat source is represented by a marker whose colour encodes the temperature level of the reported waste heat. A continuous colour scale from blue to red is applied, where blue represents low-temperature sources (below 40 °C), yellow and orange denote medium-temperature ranges (75-110 °C), and red highlights high-temperature sources above 150 °C. The marker shape is linked to the WHS category, enabling users to visually distinguish between the types of medium (see Fig. 9). When selecting an individual source, a popup window provides detailed information including company name, location, temperature, available power, and calculated distance to the nearest district heating area. This enables users to quickly assess the relevance of a source without leaving the map interface.

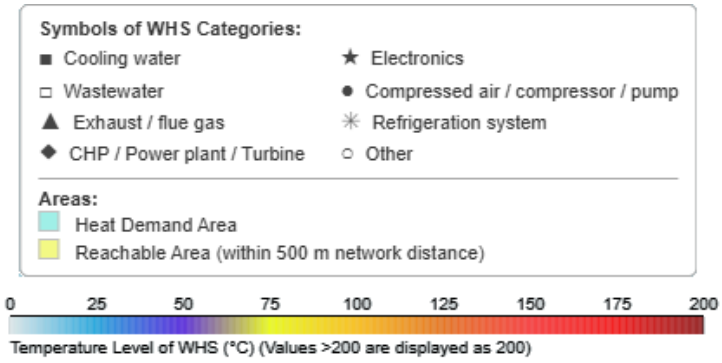


Fig. 9. Map legend.

Representation of Reachability Areas. For each industrial site, the computed reachability area - derived from the network-based proximity analysis - is displayed as a semi-transparent yellow polygon surrounding the source-marker. These polygons illustrate the area that can be reached within the defined maximum path length of 500 m along existing road networks. They provide a visual impression of potential connection corridors between the waste heat source and nearby consumer zones.

Intersection Visualisation. Areas where the reachability polygons overlap with heat demand regions are highlighted in green, indicating spatial matches between potential waste heat sources and feasible district heating areas, as shown in Fig. 7. These intersections represent zones of particular interest for further technical or economic evaluation, as they combine sufficient temperature level, available thermal power, and spatial proximity, depending on the users filter settings. The green overlay thus serves as a visual indicator of potential integration opportunities.

Map Interaction and Functionality. Users can freely pan and zoom across the map, toggle individual layers on or off and apply filter settings to display only selected categories, temperature ranges or source types. The map view automatically adjusts its extent when filters are applied, ensuring that relevant features remain visible at appropriate zoom levels. Radial fan layout ensures that multiple features with identical or near-identical coordinates are displayed without overlap (see Fig. 10)

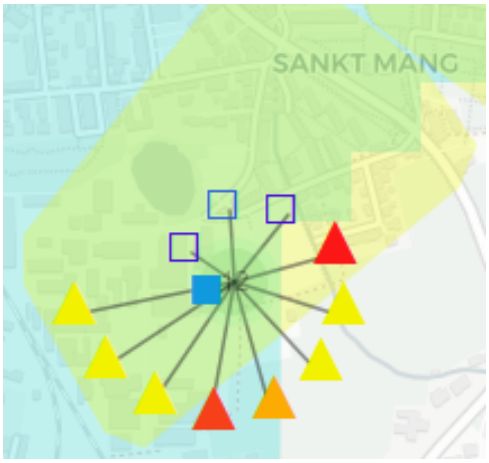


Fig. 10. Multiple WHS with different temperature levels at one industrial site.

Overall, the interactive map representation forms the core of the user experience within the webtool. It transforms numerical and spatial data into an interpretable visual environment that supports user-driven exploration, comparison, and qualitative assessment of waste heat utilisation potentials.

5. Conclusions

Large-scale heat pumps represent a promising technology for the decarbonization of the heating sector, particularly within existing DH systems. However, their deployment remains limited due to complex planning processes and a lack of early-stage planning tools.

The *HeatSHIFT Webtool* addresses this gap by providing a freely accessible, user-friendly platform for the preliminary techno-economic and environmental evaluation of large-scale heat pump systems. It supports informed decision-making by combining refrigerant selection, cycle simulation, cost estimation, scenario and sensitivity analysis, and ecological assessment within a single framework.

By relying on validated default values and simplified modeling approaches, the tool enables a structured comparison of technology options, even in the absence of detailed input data. It thus provides valuable orientation for planners, utilities, and policymakers committed to accelerating the heat transition.

The *HeatSHIFT Interactive Platform for Waste Heat Mapping and Analysis* presented here provides a structured and accessible way to screen industrial waste heat potentials by integrating national BAFA data with geospatial analysis, a transparent classification approach, and an intuitive interactive map. Through its combination of linear and network-based proximity calculations, it enables users to quickly identify where technically accessible waste heat sources coincide with areas of high heat demand.

Although the tool is not intended to perform full techno-economic assessments, it offers a valuable first step in narrowing down promising candidate sites for further investigation. By transforming heterogeneous data into a coherent, user-driven spatial exploration environment, the platform supports early-stage planning and increases transparency in the identification of waste heat utilisation opportunities.

Future enhancements may include additional infrastructure datasets, refined temporal modelling, and the development of a structured evaluation framework to further strengthen the platform's analytical capabilities.

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